

Introduction to **User Experience Design**

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Aim

What is User Experience (UX)

Why it is important

How?  Nielsen's Usability Heuristics

- Visibility of System Status

User Experience

The **User Experience** (UX) is how a user interacts with and experiences a *product*, *system* or *service*. It includes a person's *perceptions of utility, ease of use, and efficiency* ([wiki](#))

- Problem-solving: Removing friction to solve user pain points.
- Aesthetic-usability effect: Beautiful designs are perceived as more usable.



UX in Action: Food Delivery App

The Foundation

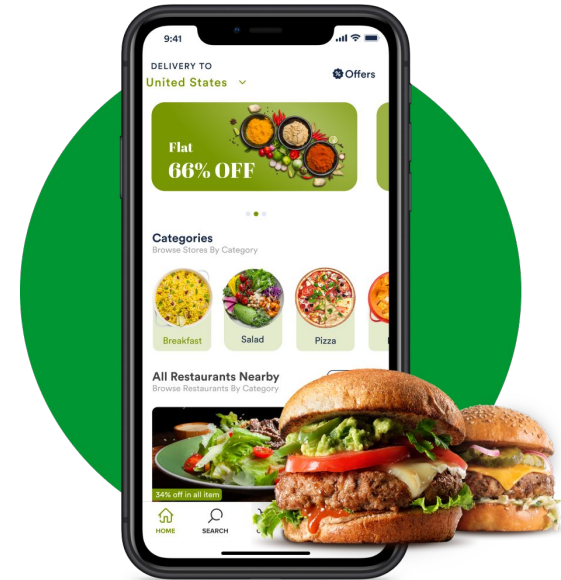
- **Utility:** Finding the right restaurant when you are hungry.
- **Ease of Use & Efficiency:** Checking out in **3** clicks; tracking your rider in real-time.

Problem-Solving

- Removing Friction: Solving the "What should I eat?" and "I don't want to go out" dilemma with smart recommendations

Aesthetic-Usability Effect

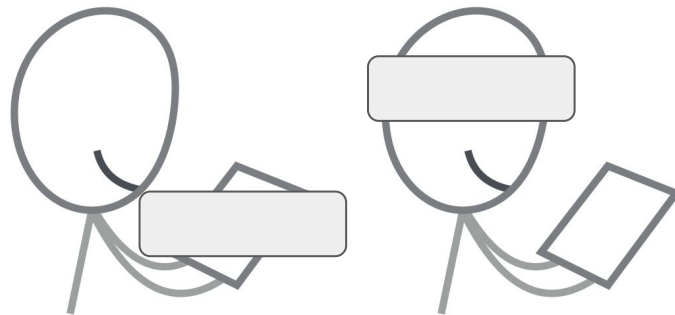
- Perceived Satisfaction: High-quality food imagery and a clean interface make the app feel faster, safer, and more reliable.



Beautiful UI Doesn't Guarantee Good UX

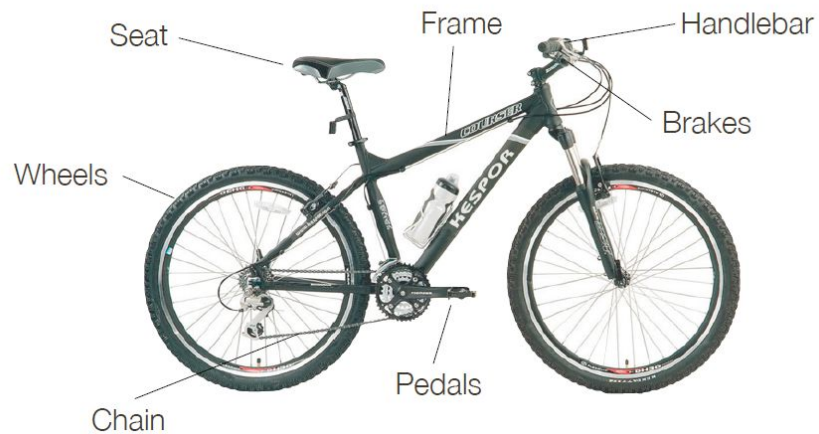
User Interface (UI) is all the external, visual aspects when you interact with a product.

User Experience (UX) is your internal cognitive and emotional response during that interaction.



<https://digitalsynopsis.com/design/graphic-design-memes/>

Expensive Doesn't Guarantee Good UX



\$ 10,000 usd



Questions (Yes or No)

Q1: If a website has no *technical* bugs and loads instantly, does that automatically mean it has a good UX?

Q2: Can we evaluate the UX of a physical product, like a kitchen microwave or a public vending machine?

Q3: An app is incredibly easy to use and beautiful, but it doesn't actually solve any problem or provide any value to the user.



Why UX is important



When encountering bad UX, you feel

**Embarrassed, Irritated, Wronged,
Displeased, Confused, Frustrated, Angry**

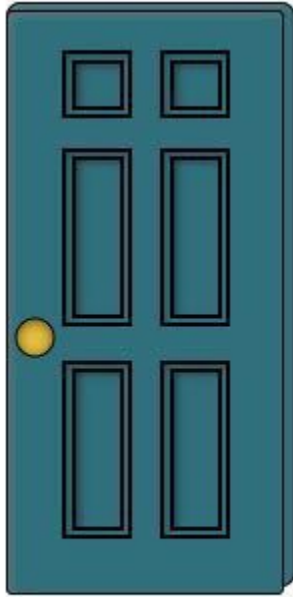


UX is Everywhere in Life

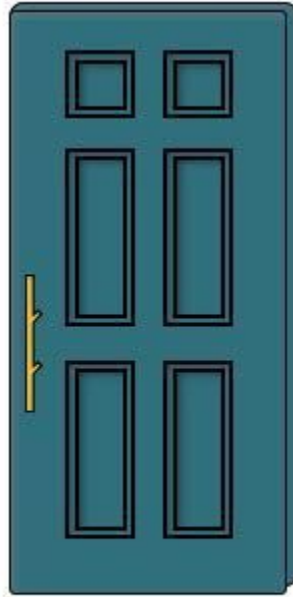


Norman's Door

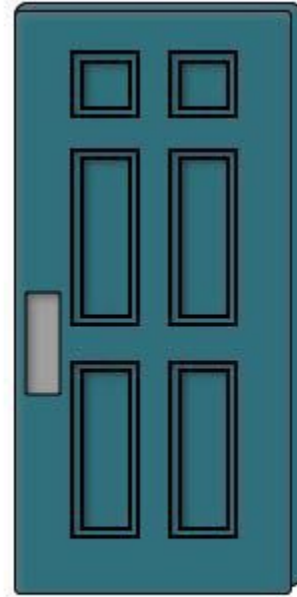
https://www.reddit.com/r/CrappyDesign/comments/5wolzl/a_norman_door_at_the_apple_store/



↻ TURN

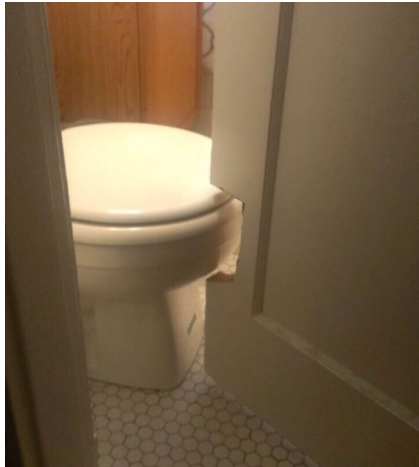


↓ PULL



↑ PUSH

<https://njkhanh.com/how-to-avoid-bad-designs-p5f32333831>

























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When you want to pre-order face masks, which one should you select?





Group discussion (skip in the EMI TP workshop)

- Share your good or bad using experience of a software system
- If you dont have one, consider [High-Speed Rail booking system](#)
- Why it is good, why it is bad
- 3-4 person per group
- 15 mins group discussion
- 5 min sharing



How to design a system with good UX?

It is NOT easy, but

Nielsen's Usability Heuristics



Jakob Nielsen: 10 UX Heuristics



Visibility Of
System Status



Match Between
System & Real
World



User Control And
Freedom



Consistency And
Standards



Error Prevention



Recognition
Rather Than
Recall



Flexibility And
Efficiency Of Use



Aesthetic And
Minimalist Design



Help Users With
Errors



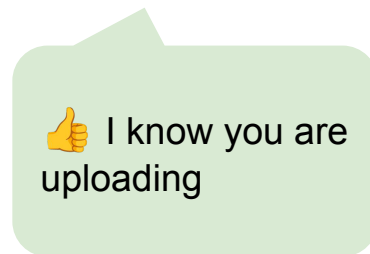
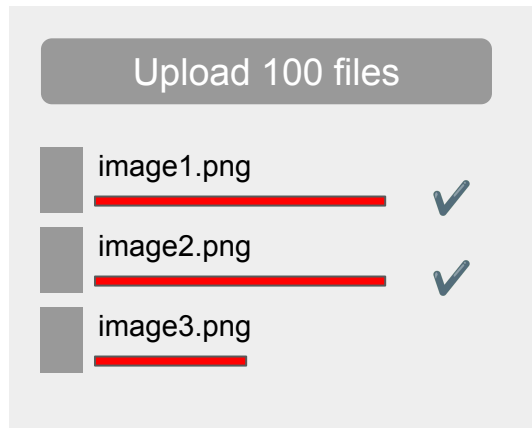
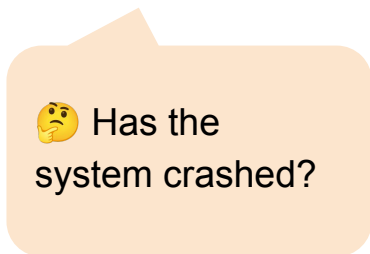
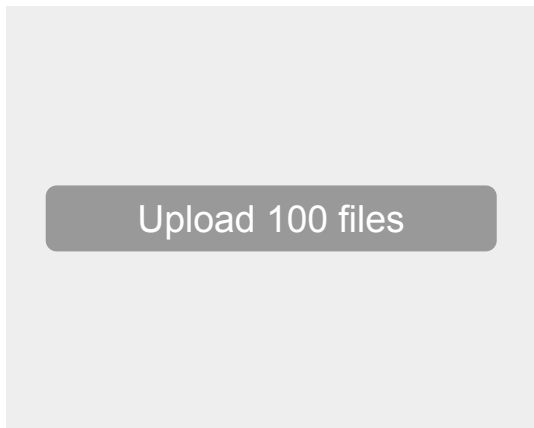
Help And
Documentation

NS01 Visibility of System Status

“The system should always keep users informed about what is going on, through appropriate **feedback** within a reasonable time.”

- Highlighting important information/elements and understanding user context to help to establish visibility.
- Providing feedback for action in a reasonable amount of time is important to retain and engage the user.







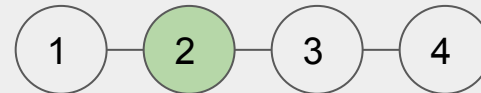
👍 Clearly
notify un-read
messages

Email

john#gml.com

invalid email

👍 Instant
warning errors



👍 Clearly state
now is at Step 2



$$75 - 37.5 = 0 \text{ ?!}$$

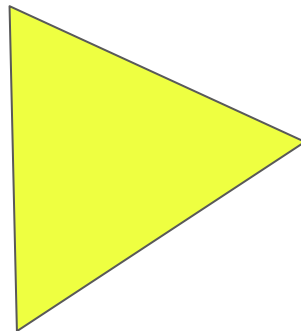
Take out your phone, open the *calculator* app, and give it a try.

Summary

UX is more than **utility** and **efficiency**; it is active **problem-solving** wrapped in an **aesthetic** package.

WHAT

HOW



WHY

No matter how powerful the features, a system with poor UX causes frustration and anger, ultimately leading to product failure.

Design Heuristics as a framework to systematically discover bugs, remove cognitive friction, and ensure system feedback.